INTRODUCTION TO ISR OPERATIONS

INSTRUCTOR'S PROFILE

Cpl Ernesto A Raydanas Jr 920672 (SC) PA
Navy Base Baguio City

Previous Unit Assignment and TO:
Charlie Company, 3IB, 7ID, PA
RTO Scout Platoon HHCo 3IB, 7ID, PA
Rembass Specialist 7MIB, 7ID, PA
IMINT Analyst ASE 7MIB, 7ID, PA
OSINT Specialist 7MIB, 7ID, PA
SIGINT Operator 7MIB, 7ID, PA
Team Leader Cyber ADO – 7MIB, 7ID, PA
Team Leader UAS Pilot 7MIB, 7ID, PA
The Intelligence School, 1st Intelligence Regiment, PA
UAS Instructor TIS, 1IR, PA

Current Unit Assignment:
Signal School, ITG, ETC, PA

Relevant Military Schooling:
CSC ctw JWMOC Class 394-15 – 7DTS, 7ID, PA
Small Unmanned Aerial Systems – GIG, ISAFP, GHQ
ASCSET/Warfighting – 7DTS, 7ID, PA
Intelligence Platoon Operations Course – 7MIB, 7ID, PA
Portable Radio Receiver Operations Course – 7DTS, 7ID, PA
Instructors Development Training – ETC, PA



INSTRUCTOR'S PROFILE



Relevant Military Schooling:
Intel Platoon Operations Course – 7MIB, 7ID, PA
Skylark 1 LEX Maintenance Course – Elbit Systems, Israel
Skylark 1 LEX Trainers Training – Elbit Systems, Israel
Technical Capability Enhancement Training – 14th AIBn, IR, PA
Agent Recruitment and Management Training – 7MIB, 7ID, PA
Swimming Trainers Training 02-25 – ETC, PA
Signal NCO Basic Course Class 136-26 – SS, ITG, ETC, PA
(with Higher Honors 94.25%)

NANG AKING BILHIN ITO AY PARISUKAT, NANG AKING BUKSAN ITO AY BILOG, AT NANG AKING KAININ ITO AT TATSULOK. ANO ITO?

Ans. Pizza

Ano ang meron sa itlog na meron din sa dagat?

Ans. SHELL

BAKIT BINUBUKSAN ANG BINTANA TUWING UMAGA?

Ans. KASI NAKASARADO

LOGIC

SUMAKAY AKO
UMUPO AKO
SUMANDAL AKO
PUMARA AKO
TUMAYO AKO
BUMABA AKO
TANONG:
ANONG DALA KO?

Rapsey

CLASSROOM RULES











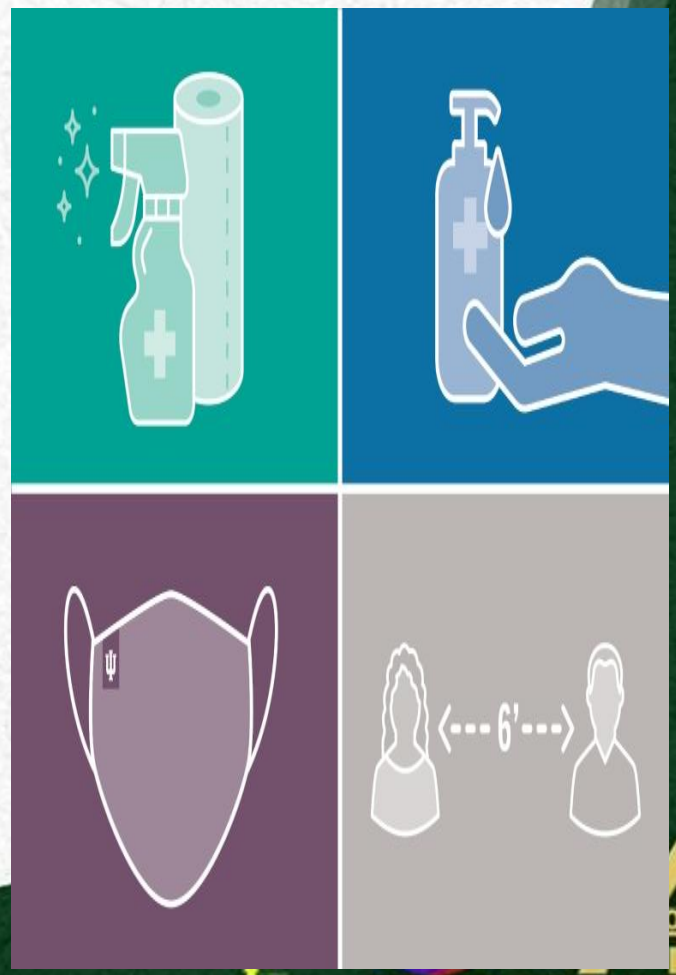


SAFETY PRECAUTIONS









NEED

ISR Operations is critical for military effectiveness, providing essential intelligence capabilities that support decision-making, enhance operational success, and contribute to mission accomplishment across various military operations and scenarios.

TIME



ROADMAP

- **Explain** the following:
  - Identify the Role and Importance of ISR. (C2)
  - Identify ISR Assets and Platforms. (C2)
  - Explain the Principles of ISR Operations. (C2)
- **Apply** (Check on learning)
- **Summarize** let the learners summarize the important topics discuss
- **Assess** through written examination

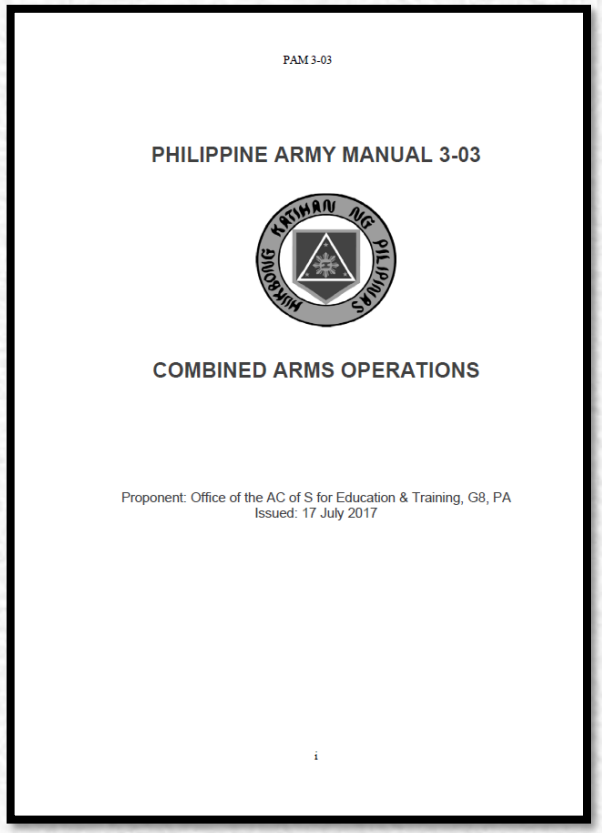


LEARNING OBJECTIVE

At the end of the subject, the students will be able to demonstrate a comprehensive understanding of Intelligence, Surveillance, and Reconnaissance (ISR) Operations. (C3)

REFERENCE

- **PAM 2 – 0103 - *UNMANNED AERIAL SYSTEM OPERATIONS dated 01 March 2022***
- **PAM 3 – 03 - *COMBINED ARMS OPERATIONS dated 17 July 2017***
- **PAM 6 – 00 - *COMMAND AND CONTROL COMMUNICATIONS, AND CYBER SYSTEMS dated 11 December 2017***

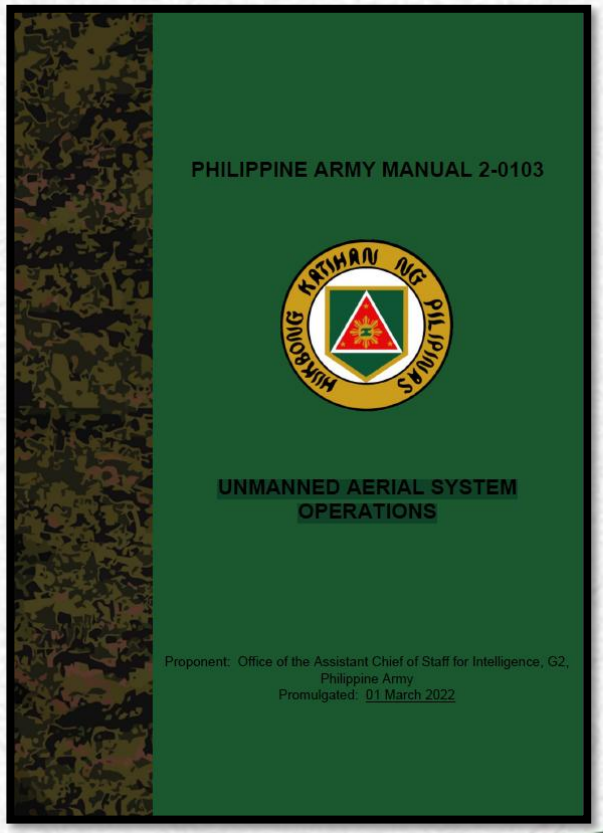

PAM 3-03

PHILIPPINE ARMY MANUAL 3-03

COMBINED ARMS OPERATIONS

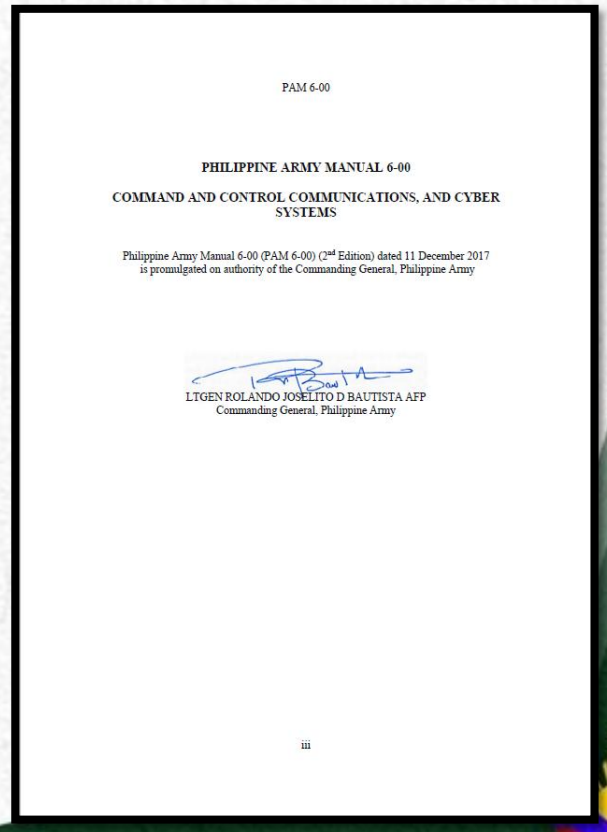
Proponent: Office of the AC of S for Education & Training, G8, PA
Issued: 17 July 2017

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PHILIPPINE ARMY MANUAL 2-0103

UNMANNED AERIAL SYSTEM OPERATIONS

Proponent: Office of the Assistant Chief of Staff for Intelligence, G2, Philippine Army
Promulgated: 01 March 2022


PAM 6-00

PHILIPPINE ARMY MANUAL 6-00

COMMAND AND CONTROL COMMUNICATIONS, AND CYBER SYSTEMS

Philippine Army Manual 6-00 (PAM 6-00) (2nd Edition) dated 11 December 2017 is promulgated on authority of the Commanding General, Philippine Army

LTGEN ROLANDO JOSELITO D BAUTISTA AFP
Commanding General, Philippine Army

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"If you know the enemy and know yourself, you need not fear the result of a hundred battles."

– Sun Tzu



ISR Operations

ISR Operations is critical for military effectiveness, providing essential intelligence capabilities that support decision-making, enhance operational success, and contribute to mission accomplishment across various military operations and scenarios.

ISR Operations



Role and Importance of ISR

Introduction

- Intelligence, surveillance, and reconnaissance (ISR) are essential components of all military operations. It provides decision-makers and action-takers with a more complete picture of the situation on the ground, in the air, at sea, in space, and cyberspace

Role and Importance of ISR

What Is Intelligence, Surveillance, and Reconnaissance (ISR)

Intelligence, Surveillance, and Reconnaissance (ISR) is an integrated operations and intelligence activity that synchronizes and integrates the planning and operation of sensors, assets, and processing, exploitation, and dissemination systems in direct support of current and future operations. It is at the intersection between military planning, operations, and assessment where intelligence is the product of surveillance and reconnaissance operations.

Role and Importance of ISR

What Is Intelligence, Surveillance, and Reconnaissance (ISR)

ISR systems can collect information from a wide range of sources, including electronic communications, optical, radar, and infrared imaging. Some of the assets used to collect this data include satellites, sensors, unmanned aircraft, aviation systems, specialized ground, sea, or space-based equipment, and human intelligence teams.

Role and Importance of ISR

What Is Intelligence, Surveillance, and Reconnaissance (ISR)

Accurate ISR data is critical for providing high-quality intelligence about enemy threats and serves to increase the effectiveness of military operations. The ongoing development of technology has only increased the demand for ISR capabilities in recent years.

Battle of Marawi, also known as the "Marawi Siege."



Role and Importance of ISR

The Components of ISR

The principles of intelligence, surveillance, and reconnaissance (ISR) have been used in warfare for centuries. Individual components of ISR include:

INTELLIGENCE: the result of surveillance and reconnaissance combined with other data;
SURVEILLANCE: the continuous observation of a target; and
RECONNAISSANCE: collecting data to address a specific military question

Role and Importance of ISR

Roles and Responsibilities of ISR

According to joint military doctrine, the primary role of intelligence is to provide information and assessments to aid mission completion.





Role and Importance of ISR

Roles and Responsibilities of ISR

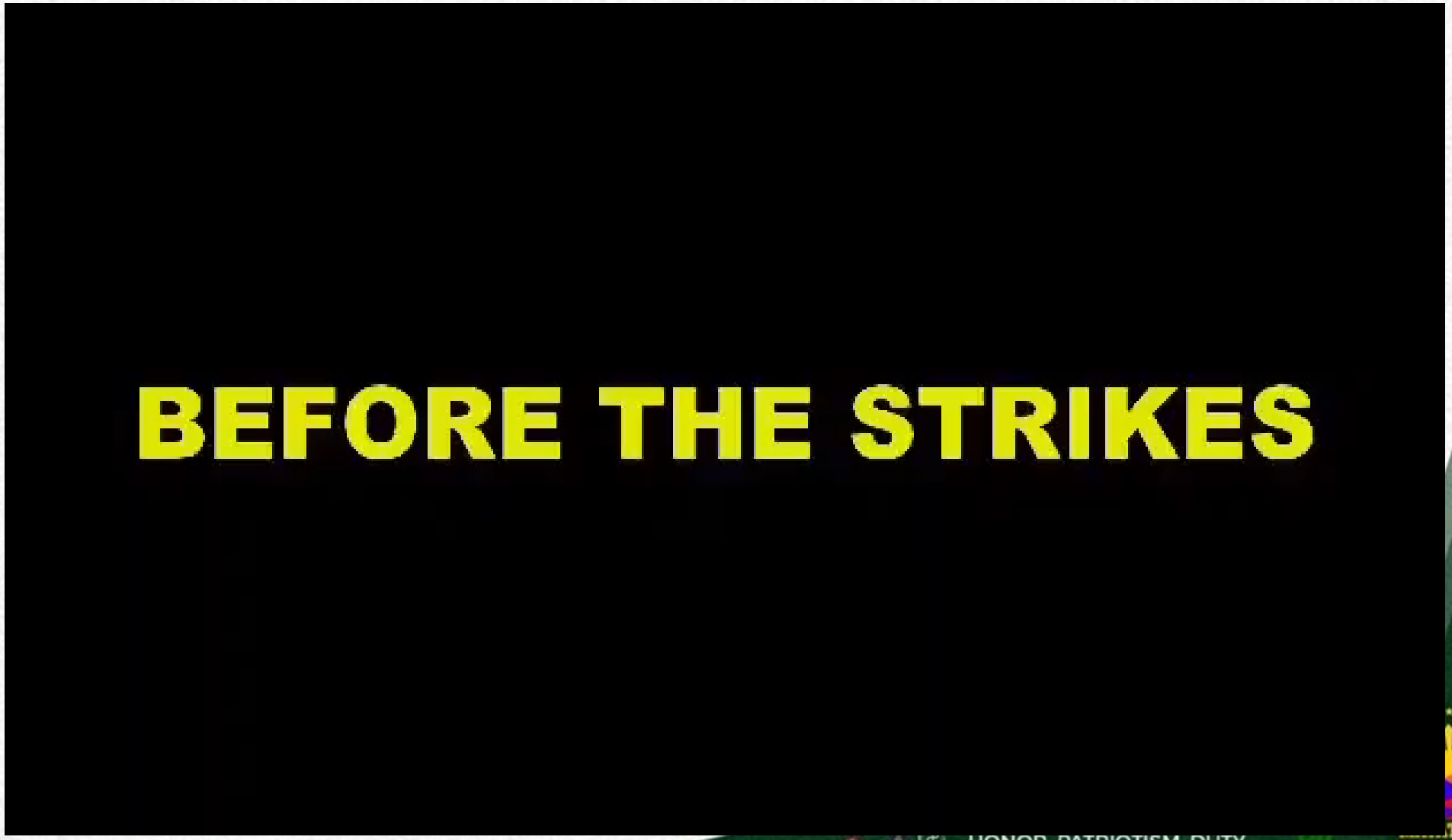
ISR aims to provide information to commanders to aid in decision-making, support military planning by anticipating enemy actions and defining the operational environment, alert friendly forces to threats, support deception techniques and counteract enemy deception, identify adversary vulnerabilities, hold targets at risk of attack, and evaluate combat effectiveness.

Role and Importance of ISR

ISR in the Military

ISR systems range in size from mobile devices to satellites. ISR systems use unstructured data to extract and analyze insights. Warfighters can make better decisions thanks to the lightning-fast information flow made possible by these insights.

Role and Importance of ISR



ISR Assets and Platforms

Airborne ISR Systems
Ground-Based ISR Systems
Maritime ISR Platforms
Space-Based ISR Systems

ISR Assets and Platforms

Airborne ISR Systems

Unmanned Aerial Vehicles (UAVs)

UAVs that carry payloads such as missiles, bombs, or precision-guided ammunitions to engage targets identified through ISR missions provide a flexible and persistent strike capability supporting ground or maritime operations.



POSSIBLE AREA: **Vicinity Kawayan, Brgy. Camam-onan, Gigaquit, SDN.**
(SDG16 & LGY, GF16, NEMRC)



Digital Evidences








CLASSIFICATION:SECRET
OBJECTIVE: 30IB
MGRS: 51PYL 99418 43138
DTG: 13 0348H SEP 20
HEAT SIGNATURES
LOCATION: OBSERVED HEAT SIG 200 M, SOUTH WEST PORTION OF T18.



ISR Assets and Platforms

Airborne ISR Systems

Manned Reconnaissance Aircraft

A more traditional approach to aerial ISR is manned reconnaissance aircraft. Manned aircraft have been used for reconnaissance since World War I and used with great effect since World War II.



ISR Assets and Platforms

Airborne ISR Systems

High-Altitude, Long Endurance (HALE) Platforms

A high-altitude, long-endurance (HALE) vehicle can operate extremely high for many hours on end since the pilot has no limitations.



ISR Assets and Platforms

Ground-Based ISR Systems

Ground Surveillance Radars

Ground surveillance radars can be either fixed, usually on a low tower, or they can be mounted to a vehicle as a mobile unit.

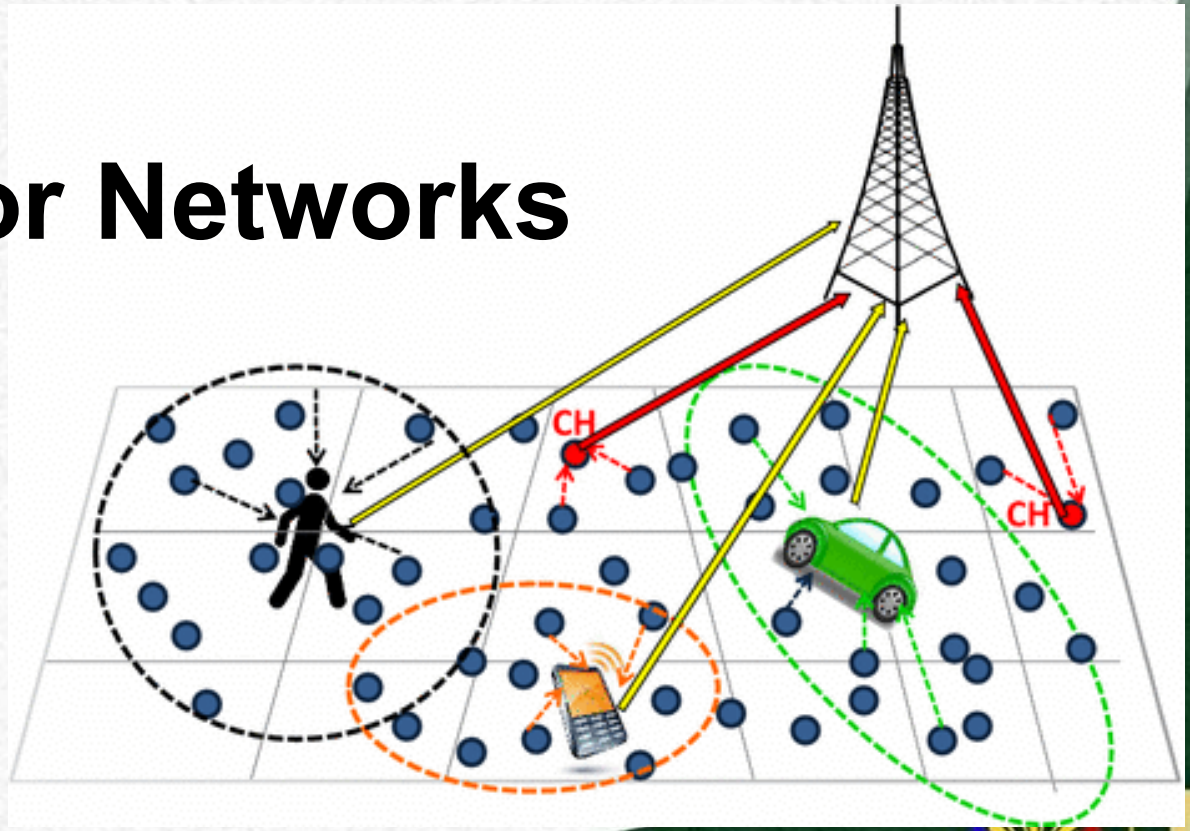


ISR Assets and Platforms

Ground-Based ISR Systems

Fixed and Mobile Sensor Networks

Sensor networks are nothing new; Allied forces have been dropping sensors behind enemy lines for decades.



ISR Assets and Platforms

Maritime ISR Platforms

Surface-Ship ISR Systems

The backbone of the Navy is its surface warfare fleet. While the aviation branches are crucial, as are submarines, it all begins with surface warfare.



ISR Assets and Platforms

Maritime ISR Platforms

Submarine ISR Systems

Submarines are silent killers. They are the most invisible asset in the entire military, which gives them an incredible ability to gather information.



ISR Assets and Platforms

Space-Based ISR Systems

IMINT (Imagery Intelligence)

IMINT is this, but far more powerful and with no information blocked out or blurred. It is incredible for observing huge areas, but since satellites are in regular orbits, targets can track these cycles and avoid being seen.



ISR Assets and Platforms

Space-Based ISR Systems

SIGINT (Signals Intelligence)

Space-based SIGINT is a tool to be used in collaboration with aerial assets and ground-based SIGINT assets.

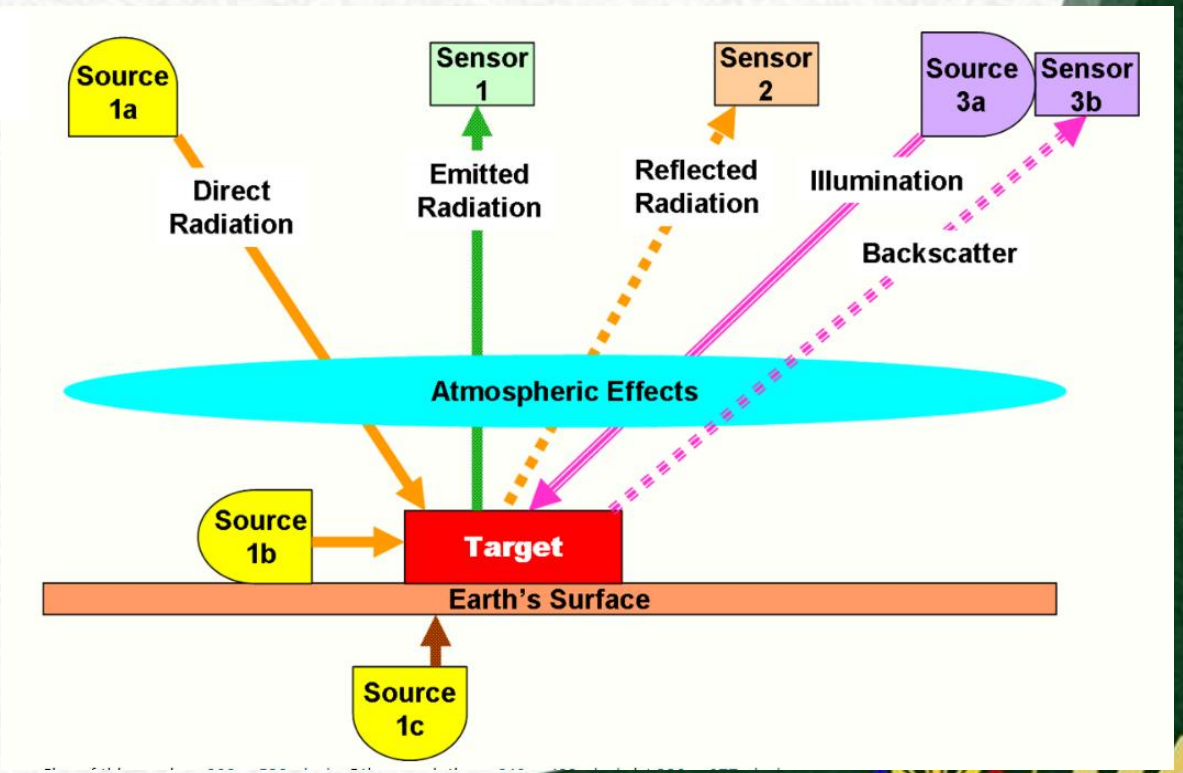


ISR Assets and Platforms

Space-Based ISR Systems

MASINT (Measurement and Signature Intelligence)

MASINT is used to identify distinctive characteristics of their targets, such as source, sender, or emitter type and location, and identify the measurement and identification of the target.

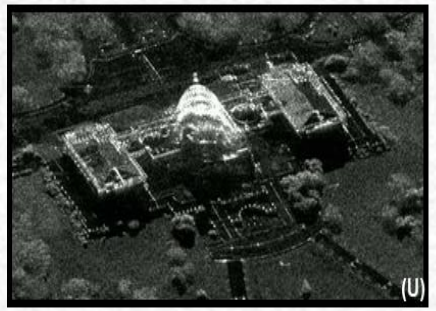


ISR Assets and Platforms

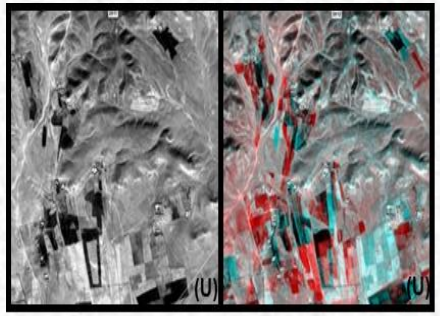
Space-Based ISR Systems

- **Benefits of space-based ISR**
  - Unrestricted and worldwide coverage, satellite sensors provide multiple types of imagery and signals collection
  - Collects in denied areas of the world where aircraft and ground intelligence cannot access
  - Beyond line-of-sight coverage of large areas of interest

- **Benefits of terrestrial-based ISR**
  - Company UAS assets can provide Near-Real-Time full-motion video relay
  - Local commander approval for relocation and tasking
  - Deployment of lethal or non-lethal effects after target identification



SAR



Change Detection



Full Motion Video



Damage areas compared to the healthy vegetation (MSI)



Image using EO

Image using SWIR



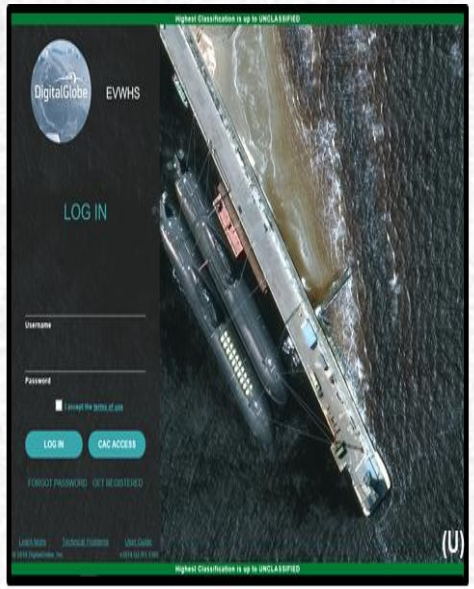
Full Motion Video

ISR Assets and Platforms

Space-Based ISR Systems

- **Benefits of commercial space-based imagery**
  - Easily obtained from a variety of sources
  - Commercial imagery products are unclassified and benefit multinational partners
  - U.S. and international companies acquire and disseminate commercial satellite imagery which can enhance intelligence capabilities
  - Commercial imagery is the easiest ISR capability to access and the most relevant for operations
    - Army Geospatial Center: https://www.agc.army.mil ~ CAC-enabled (Engineer Route Studies, Common Map Background, Urban Tactical Planners)
    - Protected Internet eXchange (PiX): https://www.pixtoday.net ~ in-depth global awareness of people, places, activities, and infrastructure
    - National Geospatial-Intelligence Agency (NGA) Map of the World: https://map.nga.mil ~ CAC-enabled
    - Army Mapping Resources: https://www.agc.army.mil/maps

Commercial Imagery Resources



https://evwhs.digitalglobe.com/myDigitalGlobe/
(MAXAR Global-Enhanced GEOINT Delivery)



SpyMeSat -
0.50m resolution



DigitalGlobe -
0.31m resolution

ISR Assets and Platforms

Space-Based ISR Systems

MAXAR Global-Enhanced GEOINT Delivery (G-EGD)





CHECK ON LEARNING

- **What are the ISR Assets and Platform**

  - **Airborne ISR Systems**
  - **Ground-Based ISR Systems**
  - **Maritime ISR Platforms**
  - **Space-Based ISR Systems**

Principles of ISR Operations

Timeliness

ISR operations prioritize the timely collection, processing, and dissemination of intelligence to ensure that commanders have the most current information available for decision-making. Timeliness allows for proactive responses to evolving situations on the battlefield.

Principles of ISR Operations

Accuracy

The accuracy of intelligence gathered through ISR assets is critical. It ensures that commanders have reliable information upon which to base their decisions, minimizing the risk of errors and misinterpretations that could lead to adverse outcomes.

Principles of ISR Operations

Fusion

ISR operations often involve multiple sources of intelligence, such as signals intelligence (SIGINT), human intelligence (HUMINT), imagery intelligence (IMINT), and others.

Principles of ISR Operations

Coverage

Effective ISR operations strive to achieve comprehensive coverage of the area of interest, whether it's a specific geographic region or a particular operational objective.

Principles of ISR Operations

Persistence

Persistence refers to the sustained and continuous monitoring of the operational area over time. ISR assets maintain vigilance to detect changes, movements, and activities that may impact mission success.

Principles of ISR Operations

Flexibility

ISR operations must be flexible to adapt to changing operational requirements and priorities. This includes the ability to quickly redirect assets, adjust collection priorities, or employ different intelligence methods based on evolving mission needs or enemy actions.

Principles of ISR Operations

Security

ISR operations involve sensitive information and assets that require strict security measures to protect against compromise or exploitation by adversaries.

Principles of ISR Operations

Interoperability

ISR assets often operate in joint and coalition environments where coordination and interoperability among different military services and allied forces are essential.

Principles of ISR Operations

Ethical Considerations

Ethics play a significant role in ISR operations, guiding the conduct of intelligence collection, analysis, and dissemination in accordance with international laws, rules of engagement, and moral principles.

CHECK ON LEARNING

- **What are the Principles of ISR Operations**
  - Timeliness
  - Accuracy
  - Fusion
  - Coverage
  - Persistence
  - Flexibility
  - Security
  - Interoperability
  - Ethical Considerations

SUMMARY

- ✓ Role and Importance of ISR
- ✓ ISR Assets and Platforms
- ✓ Principles of ISR Operations

LEARNING OUTCOME

By this time, students were able to discuss the concept of Intelligence, Surveillance, and Reconnaissance (ISR) Operations.

FEEDBACK





FUTURE

Signal Support in Combined Arms Operations

THANK YOU!



HAVE A NICE DAY